

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1278

Twinbrook 345 kV
September 03, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1278. The Interconnection Customer proposes a 200.3 MW solar plus storage facility interconnecting at the Twinbrook 345 kV substation in the Maryland area. The request was received on August 30, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 333 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,401,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
LINDEN – KENSINGTON 138 KV	KENSINGTON – LINDEN SUBSTATION 138 KV	144.8	321	5.7
KENSINGTON – LINDEN 138 KV	KENSINGTON – LINDEN 138 KV Line	138.5	335	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
KENSINGTON – LINDEN 138 KV #2	Rebuild line with higher-capacity conductor	\$1,403,000	88% (\$1,229,000)
KENSINGTON – LINDEN 138 KV Circuit 1	Rebuild line with higher-capacity conductor	\$1,338,000	88% (\$1,172,000)
	Total allocated to project		\$2,401,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.